IES · GenAI for Engineers

Lab 3.2 · Prompt
DMAIC Improvement Report

How this prompt is built

This prompt follows **R.I.S.E.** and adds **E-MAP** for engineering context — **E**ngineering context, **M**etrics, **A**ssumptions and **P**arameters. E-MAP is what stops a generic answer: it pins the units, the assumptions that must stay visible, and the boundary of the data in scope.

	Block	What it does
R	**Role**	Sets the perspective Copilot answers from.
I	**Input**	Names exactly which evidence may be used, and forbids the rest.
S	**Steps**	Makes the work traceable so a reviewer can follow it.
E-MAP	**Engineering context, Metrics, Assumptions, Parameters**	Pins the units, the assumptions that stay visible and the data in scope.
E	**Expectations**	Defines what finished looks like, and what must never appear.
YOU	**Independent check**	Your hand calculation. This one is not Copilot's job.

Every block below is labelled with its framework letter. Keep the labels when you reuse this prompt — they are what make it repeatable.

Copy this prompt into Copilot in Excel

Select the named table first, then open Copilot and paste everything below.

R ROLE · Role

```
Act as a Six Sigma project assistant preparing a draft for a competent reviewer.
```

I INPUT · Input

```
Use only the named Excel table WeeklyData and the category counts on the Pareto sheet.
Three weeks are Baseline and three are Pilot. Do not use outside knowledge.
```

S STEPS · Steps

```
1. Calculate total opportunities, total deviations and the deviation rate for each phase,
using deviations divided by opportunities. Show each formula and its result.
2. State the absolute change in percentage points between the phases.
3. Build the Pareto of baseline deviation categories with a cumulative percentage column
and insert the Pareto chart.
4. Draft a five-row DMAIC summary: Define, Measure, Analyse, Improve, Control.
5. In Analyse, separate what the data shows from what is only a hypothesis.
```

E-MAP ENGINEERING CONTEXT (E-MAP) · Engineering context, Metrics, Assumptions, Parameters

```
Engineering context: a chemical dosing skid whose delivered dose is checked weekly.
Metrics: deviation rate as deviations divided by opportunities, in percent, to two decimals.
Assumptions: the deviation definition is unchanged across all six weeks and every week
was counted the same way. State this assumption in the output.
Parameters: six weeks only, three per phase. No other period is in scope.
```

E EXPECTATIONS · Expectations

```
Show every formula, unit and row count so a reviewer can reproduce the numbers by hand.
State plainly that a three-week pilot does not prove the cause of the change and that two
concurrent changes cannot be separately attributed. Do not invent a root cause, a cost saving
or a completion percentage. Return one next evidence check.
End with DRAFT FOR REVIEW.
```

YOU INDEPENDENT CHECK · Independent check

```
Verify the baseline rate by hand: 36 divided by 400 = 9.00 percent.
Verify the pilot rate: 7 divided by 200 = 3.50 percent. The absolute change is 5.50
percentage points, not a 61 percent improvement claim.
```

All records are fictional training data for classroom use. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use any output to operate, maintain or approve real equipment.